SPAX: Fully Sparse Framework With Hierarchical Spatio-Temporal Fusion for Moving Object Tracking in Satellite Videos

Zhehao Xiao, Fang Xu, Chuandong Liu, Wen Yang, *Senior Member, IEEE*, and Gui-Song Xia, *Senior Member, IEEE*

***Abstract*—Moving object tracking is a fundamental task in video satellite technologies. Remote sensing scenes feature small objects and large background ratios in spatial dimension, leading existing methods characterized by dense computation to incur considerable unnecessary computational overhead of redundant regions. Moreover, the coupled motion between the satellite platform and ground objects introduces temporal complexity that current methods find difficult to achieve accurate dynamic perception. To address these issues, we propose a fully sparse framework with hierarchical spatio-temporal fusion (SPAX). Specifically, SPAX utilizes an object-centric fully sparse paradigm to reduce computational redundancy by focusing only on foreground regions. Furthermore, we adopt the hierarchical spatio-temporal fusion (HSF) to address the complexity of dual-motion coupling through intraframe multiscale feature fusion, interframe symmetric feature interaction, and interframe asymmetric feature interaction (IAFI), thereby enabling comprehensive temporal information utilization. In addition, we propose a plug-and-play Gaussian-based trajectory association (GTA) strategy to mitigate the negative impact of observational drifts and accumulated errors. Experiments show that SPAX outperforms previous methods on two popular benchmarks, achieving notable improvements of 5.1 and 7.6 on multiple object tracking accuracy (MOTA). While achieving the state-of-the-art (SOTA) performance, SPAX reduces GFLOPs by 88.4% and delivers a 2.7× speedup on SatVideoDT dataset, along with 93.2% GFLOPs reduction and up to a 3.1× acceleration on SatMTB-MOT dataset compared to our baseline. Furthermore, SPAX-Light outperforms the previous SOTA method by 6.6 MOTA and runs at 5.9× its inference speed on SatMTB-MOT dataset. Our project page: https://lebron-2016.github.io/SPAX-page**

***Index Terms*—Fully sparse framework, hierarchical spatio-temporal fusion (HSF), moving object tracking, satellite video.**

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Zhehao Xiao and Chuandong Liu are with the School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: xiaozh@whu.edu.cn; chuandong.liu@whu.edu.cn).

Fang Xu is with the School of Artificial Intelligence, Wuhan University, Wuhan 430072, China (e-mail: xufang@whu.edu.cn).

Wen Yang is with the School of Electronic Information and the School of Artificial Intelligence, Wuhan University, Wuhan 430072, China (e-mail: yangwen@whu.edu.cn).

Gui-Song Xia is with the School of Computer Science and the School of Artificial Intelligence, Wuhan University, Wuhan 430072, China (e-mail: guisong.xia@whu.edu.cn).

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I. Introduction

With the rapid development of remote sensing satellite technology [1], [2], [3], the scope of remote sensing information processing has expanded from static image analysis to the study of dynamic scenes [4], [5], [6]. Intelligent interpretation of satellite videos has become an urgent need and represents a new frontier in remote sensing [3]. Moving object tracking, which focuses on locating and tracking dynamic ground objects [7], plays a crucial role in a wide range of real-world applications, including military surveillance, traffic planning, and smart city development [7], [8], [9]. However, research in this field remains underexplored, particularly with regard to two key issues:

1) Background Redundancy in Spatial Dimension: Remote sensing images typically have large size, while moving objects are usually quite small [1], [6], resulting in a large proportion of background regions [e.g., 99.8% in SatVideoDT dataset [10] and 99.5% in SatMTB-MOT dataset [6], as shown in Fig. 1(a)]. Processing these redundant areas at high resolution imposes substantial computational and time costs, which adversely affects the efficiency.

2) Dual-Motion Coupling in Temporal Dimension: Moving elements in satellite videos not only include the smooth movements of ground objects, but also the uneven motion of the satellite platform itself, which leads to irregular pixel-level displacements and dynamic intensity variations between consecutive frames [as shown in Fig. 1(b) and (c)]. Due to the appearance degradation of small moving objects, their accurate perception demonstrates a strong temporal dependence. However, the combination of the two distinct motion systems increases the difficulty of temporal modeling, thereby raising a higher demand for effective temporal information utilization in this task.

Moving object tracking algorithms in satellite videos can be categorized into two types: tracking by detection (TBD) [11], [12], [13] and joint detection and tracking (JDT) [5], [14], [15]. As shown in Fig. 1(d), existing methods of both paradigms often employ dense feature computations to capture spatial information of objects within the scene [11], [12], [15], [16]. Although these approaches achieve remarkable performance, they also incorporate a significant amount of redundant background into feature computation, which contributes little to moving object perception and introduces unnecessary time

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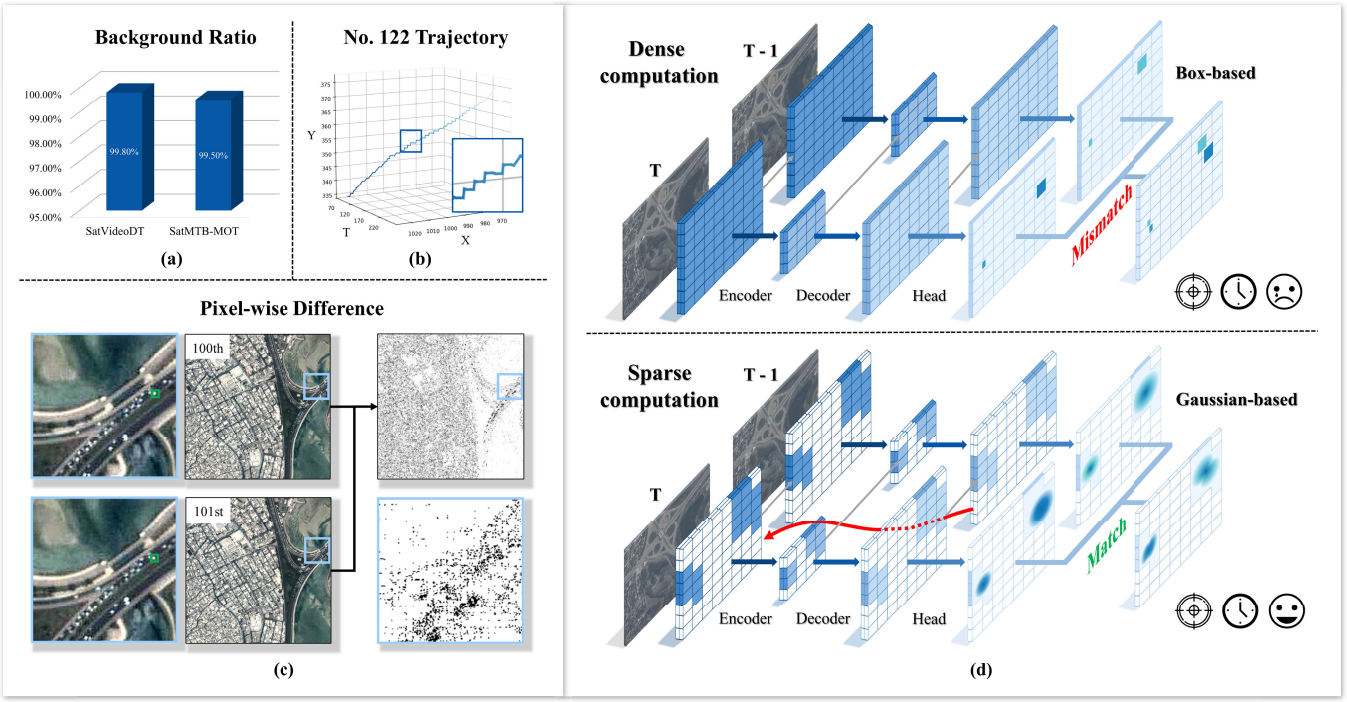


Fig. 1. (a) Background occupies 99.8% of the SatVideoDT [10] dataset and 99.5% of the SatMTB-MOT [6] dataset. (b) Position label for trajectory No. 122 in SatVideoDT-001 reveals the irregular pixel displacements. (c) Grayscale difference between the 100th and 101st frames in SatVideoDT-001 reveals the dynamic intensity variations. (d) Compared with the existing paradigm (top), our proposed SPAX (bottom) achieves fully sparse computation in the spatial dimension and introduces asymmetric feature fusion (red lines) in the temporal dimension, leading to dual benefits in both effectiveness and efficiency. In addition, the Gaussian-based object modeling enables more stable and robust tracking.

consumption. Some generic trackers [17], [18] gain efficiency with large downsampling factors. However, when dealing with diminutive objects in this task, such a process leads to the loss of a crucial feature, resulting in significant performance degradation. Furthermore, most existing related methods [11], [14], [15], [19], [20] are based on concepts from the general tracking domain, employing simple interframe symmetric feature fusion in the temporal dimension [the gray connections in Fig. 1(d)]. And they struggle to effectively model the critical motion information arising from the combined movements of the satellite and objects.

We attribute the aforementioned two issues to a mismatch in computational resource allocation: redundant spatial feature computation and insufficient temporal utilization. To address these challenges, we propose a fully sparse framework with hierarchical spatio-temporal fusion, called SPAX. Our SPAX comprises two key components: the object-centric fully sparse paradigm and the hierarchical spatio-temporal fusion (HSF). Specifically, the object-centric fully sparse paradigm leverages motion priors to obtain a foreground proposal mask, computing only within the activated regions during inference, effectively eliminating the computational redundancy introduced by the background. Note that we implement custom CUDA kernels for our unique sparse feature computation operations, building upon the previous method [21]. HSF achieves comprehensive spatio-temporal feature integration based on intraframe multiscale feature fusion, interframe symmetric feature interaction, and interframe asymmetric feature interaction (IAFI). Through powerful temporal modeling and enhanced dynamic information representation capabilities, this framework robustly deals with the interference from a dual-motion coupled environment.

In addition, considering the disruption of box-based ranging caused by high-frequency satellite disturbances and the impairment of Kalman filter [22] estimation due to accumulated motion errors, we propose a Gaussian-based trajectory association (GTA) strategy. This strategy outperforms the traditional box-based strategy, significantly enhancing tracking stability and robustness. Moreover, our GTA is plug-and-play, which allows other trackers to easily benefit from it.

To sum up, our contributions are presented as follows.

1) We propose an SPAX, which aims to comprehensively address the challenges posed by spatial redundancy and temporal complexity. To the best of our knowledge, this is the first work to fully integrate feature-level sparsity into deep networks for satellite video analysis.
2) We propose HSF to enhance dynamic feature representation, thereby improving the model's ability to locate moving objects while ensuring perceptual stability in complex scenarios.
3) We design a plug-and-play GTA strategy to overcome uniform distribution limitations in a dual-motion coupled system.
4) Extensive experiments on the SatVideoDT and SatMTB-MOT datasets demonstrate that, while achieving the state-of-the-art (SOTA) performance, SPAX reduces GFLOPs by **88.4%** and achieves a **2.7**× speedup on the SatVideoDT dataset, along with a **93.2%** reduction in

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GFLOPs and up to a **3.1**× acceleration on the SatMTB-MOT dataset compared to the baseline. Furthermore, SPAX-Light outperforms the previous SOTA method by 6.6 MOTA and runs at **5.9**× its inference speed on SatMTB-MOT dataset.

II. Related Work

A. Multiobject Tracking

Common multiobject tracking algorithms can be categorized into two types: TBD and JDT. The TBD paradigm utilizes a detector to obtain candidate boxes, followed by an association algorithm to generate object trajectories. With the rapid development of object detection, numerous superior detectors [23], [24], [25], [26], [27], [28] have been integrated into tracking tasks. Combined with simple yet efficient association methods [17], [29], [30], this paradigm has been widely adopted in various domains. In comparison, the JDT paradigm [16], [18], [31], [32] employs a shared model to simultaneously perform object detection and feature extraction. This allows for robust association by combining motion and appearance information.

Although existing generic trackers have achieved impressive performance on common video datasets, their effectiveness is significantly degraded when applied to satellite videos, which are characterized by small objects with weak appearance features. This limitation is further exacerbated by their large downsampling ratio, which inevitably leads to critical feature loss of small objects. Moreover, most general trackers rely on single-frame or adjacent-frame implementations, failing to fully exploit temporal dynamic information, which is vital for estimating the motion attribute of objects. As a result, they struggle to meet the demands of moving object tracking in remote sensing scenarios.

B. Moving Object Detection and Tracking

The task of detecting and tracking moving objects in satellite videos focuses exclusively on dynamic objects, imposing additional requirements on conventional detectors and trackers, particularly in terms of temporal strategies. Considerable researches have been conducted on moving object detection. Traditional methods [8], [33], [34], [35], [36], [37] leverage interframe differences and optical flow, combined with convex optimization and handcrafted rules to separate moving foreground from background. However, these methods struggle to extracting dynamic regions and fall short of achieving higher level image interpretations, such as object class prediction. With the advancement of deep learning, recent works have focused on temporal interaction and fusion at the feature level [5], [11], [12], [14], [15], [20], [38], [39], [40], demonstrating higher accuracy. Among them, representative works following the TBD paradigm include CKDNet-SMTNet [11], DSFNet [12], and Hieum [13], while those based on the JDT paradigm include CFTracker [14], TGraM [5], and MP^{2}Net [15]. Specifically, DSFNet [12] employs stacked 3-D convolutions to capture dynamic video features. CFTracker [14] uses an attention mechanism to fuse shallow and deep features between adjacent frames. MP^{2}Net [15] leverages independent ConvGRU modules to perform multiframe temporal fusion, achieving great performance.

However, most existing solutions for moving object perception are constrained within frameworks that focus on multiscale interaction within the same time step or symmetric feature interaction across time steps. These approaches overlook cross-frame asymmetric information exchange, limiting the full exploitation of temporal information. Furthermore, the computationally intensive network poses significant challenges to algorithmic efficiency.

C. Sparse Networks

In recent years, sparse convolution [41], [42] has emerged as a promising approach for accelerating neural network inference. Unlike model pruning, which focuses on trimming model structures, sparse networks operate primarily at the feature level, computing only specific regions in the feature space while skipping the rest. Due to the inherently sparse nature of point cloud distributions, 3-D sparse networks based on point clouds have seen rapid development and maturation in recent years [43], [44]. In the 2-D domain, QueryDet [45] employs a cascaded sparse architecture to accelerate small-object detection. CEASC [46] predicts the activation mask ratio to enable sparse computation in the detection head, while ESOD [47] further extends the scope of sparse network to the neck. In addition, though some approaches [21], [48], [49], [50], [51] apply sparse computation to process videos, they face two critical issues: additional training modules that increase system overhead, and inherent limitations in handling interference from complex scenarios like nonuniform camera motion.

Compared to prior works, we propose a cost-free, fully sparse network that leverages motion priors and a mask consistency strategy for object-centric mask updating, enabling its effective application in challenging satellite videos.

III. Method

A. Overview

The overall structure of SPAX is shown in Fig. 2. Let $\boldsymbol{I} = \{\boldsymbol{I}_t \in \mathbb{R}^{H\times W\times 3}, t = 1, 2, \ldots, n\}$ denotes a video sequence, consisting of n frames. For the image $\boldsymbol{I}_t$ at time t, we incorporate its historical m frames to derive motion priors, generating the proposal mask $\widetilde{\boldsymbol{M}}_t$. Using k frames as the network input, along with $\widetilde{\boldsymbol{M}}_t$, we perform fully sparse computation within the network. First, the frames are processed through an encoder to obtain static feature sets $\{\boldsymbol{C}_i\}_{i=t-k+1}^{t}$. Subsequently, these static feature sets are fed into the HSF framework. Based on the deep static feature sets $\{\boldsymbol{F}_i\}_{i=t-k+1}^{t}$ produced by intraframe multiscale feature fusion, interframe symmetric feature interaction and IAFI are conducted, generating feature sets $\{\boldsymbol{P}_i\}_{i=t-k+1}^{t}$ that encapsulate both static and dynamic information. Finally, the decoder decodes $\boldsymbol{P}$ to output heatmap sets $\{\boldsymbol{H}_i\}_{i=t-k+1}^{t}$, object-size sets $\{\boldsymbol{S}_i\}_{i=t-k+1}^{t}$, center-offset sets $\{\boldsymbol{O}_i\}_{i=t-k+1}^{t}$, and predicted-mask sets $\{\boldsymbol{M}_i^{\text{res}}\}_{i=t-k+1}^{t}$. The predicted mask $\boldsymbol{M}^{\text{res}}$ at each time step will impose a consistency strategy for proposal mask generation at the subsequent time step. The GTA strategy leverages the Gaussian representation of objects to achieve

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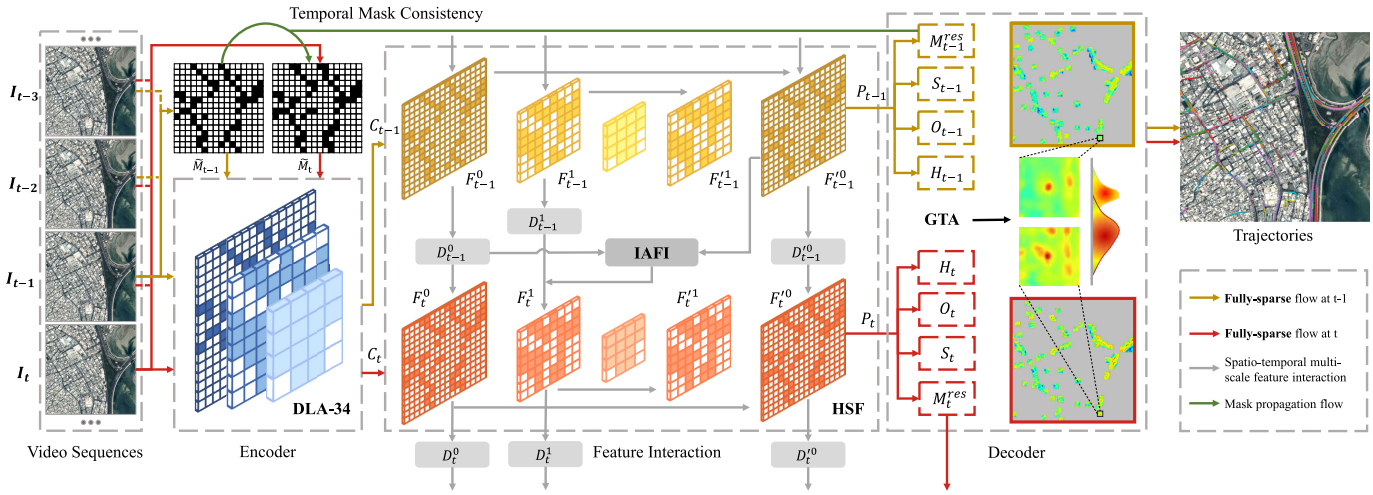


Fig. 2. Overall structure of SPAX. First, we generate the proposal sparse mask for the current frame using historical frames and masks. Then, the current image and the proposal mask are fed into the network for fully sparse computation. The network architecture consists of three main components: 1) an encoder to capture the static features of the image; 2) an HSF framework to extract rich spatio-temporal information; and 3) a decoder to decode the extracted features to generate the final results, which utilizes the GTA strategy to ensure robust object association and trajectory generation.

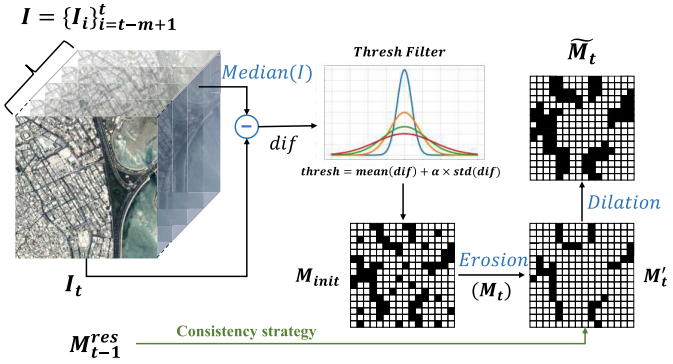


Fig. 3. Illustration of the proposal mask generation process.

Algorithm 1 Proposal Mask Generation

Input: Frame set $I = \{I_i\}_{i=t-m+1}^{t}$; Current frame I_t; The predicted mask from previous time step M_{t-1}^{res}; Mask threshold α; Region radius r; Consistency threshold s.

Output: Proposal mask $\widetilde{M}_t$.

$dif = abs(I_t - median(I))$.
$\mu = mean(dif)$, $\sigma = std(dif)$.
$thr = \mu + \alpha * \sigma$.
$M_{init} = activation(dif - thr) > 0$.
$M_t = minpool(M_{init})$. // Remove outliers.
$M'_t = M_t \mid (M_{t-1}^{res} > s)$. // Consistency constraint.
$kernel = ones(kernel_size = (2 * r + 1))$.
$\widetilde{M}_t = conv2d(M'_t, kernel)$.
return $\widetilde{M}_t$.

robust association, generating continuous motion trajectories based on the initial decoding results.

B. Object-Centric Fully Sparse Paradigm

Satellite imagery is characterized by large scale and small object size. This inherent property results in a significant background redundancy in the spatial dimension, thereby adversely affecting the efficiency of the algorithm. To address this issue, we propose a fully sparse paradigm, where each layer of the backbone, neck, and head only computes the specific regions of the image based on the proposal mask. This approach consists of two key components: object-centric proposal mask generation and fully sparse computation of the network. The second component is implemented using our unique feature-level sparse operators. In the following, we will focus on detailing the first component.

A major challenge lies in obtaining the interested regions before executing the entire network. However, for moving object tracking, this issue can be efficiently resolved by utilizing temporal information from the video sequence and motion priors of the objects. Xiao et al. [13] has attempted to address this by converting small object into point cloud representation, but this approach is prone to information loss of object itself and contradicts domain expertise—successfully perceiving small objects necessitates the use of enough background information [52]. Based on the above analysis, we propose a more adaptable foreground mask extraction scheme that activates the object itself and its surrounding background regions, retaining essential information while ensuring sparsity. In addition, considering the nature of the tracking task, we introduce a consistency strategy for the proposal mask to ensure the computational continuity, which uses a mask prediction head in the decoder to offer reliable regions within the current activated parts and propagate them to subsequent frames. The detailed process is illustrated in Fig. 3 and Algorithm 1.

To calculate the proposal mask $\widetilde{M}_t$, we take frame set $I = \{I_i\}_{i=t-m+1}^{t}$, current frame I_t and the predicted mask M_{t-1}^{res} from previous time step as inputs, along with the mask threshold α and region radius r. First, we treat the absolute difference between I_t and the median of I as the differential result dif. Next, we apply a thresh filter to dif, using its mean, standard deviation and a hyperparameter α to obtain M_{init}

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[13]. Due to displacements in the image plane, M_{init} inevitably contains a significant amount of noise, i.e., outliers. Therefore, we apply an erosion operation using minimum pooling to remove the outliers, resulting in M_t. Then, we impose the consistency strategy on M_t, using the mask M_{t-1}^{res} predicted by the decoder at time $t-1$ to retain the regions with high response values from the previous frame. Finally, we perform a dilation operation on the active regions in M_t' by using a kernel of radius r filled with ones, which ultimately produces the proposal mask $\widetilde{M}_t$ for the fully sparse feature computation. The dilation operation effectively restores objects' boundary details while simultaneously enriching the surrounding contextual information. Notably, the entire method in this section is cost-free, without adding training parameters or costs.

C. Hierarchical Spatio-Temporal Fusion

In satellite videos, there is a dual-motion coupled phenomenon resulting from the movement of ground objects and the uneven motion of the satellite platform. On the one hand, this combination of dual dynamics complicates the decoupling of motion cues, thereby interfering with the assessment of object motion properties. On the other hand, typical moving objects in satellite imagery, such as vehicles and ships, occupy very few pixels and suffer from severe appearance information loss, making their accurate perception highly temporally dependent [12], [15]. As a result, effectively utilizing temporal information is in higher demand for this task. However, existing methods still have limitations in their temporal strategies. Primarily relying on interframe symmetric feature fusion, these methods are constrained to same-level interactions with limited representational capacity, consequently failing to effectively model the critical motion information in a dual-motion coupled environment.

To address this issue, we propose an HSF framework, which consists of three components: intraframe multiscale feature fusion, interframe symmetric feature interaction, and IAFI. The first one employs structures such as feature pyramid networks [53] and deep layer aggregation [54] to facilitate feature aggregation within a single time step (as shown by the horizontal gray arrows in Fig. 2 of the HSF framework). The second one extracts intralayer motion patterns via same-level feature interactions, while the third one extends this by introducing interlayer cross-frame interactions, enabling shallow features to surpass representational limits and integrate deep semantic information, in order to generate more distinctive and robust dynamic representations.

1) Interframe Symmetric Feature Interaction: In this section, we utilize the deep static feature sets $\{\boldsymbol{F}_i\}_{i=t-k+1}^{t}$ generated by intraframe multiscale feature fusion to perform temporal interaction of symmetric features across frames. Specifically, for each layer of the features in $\boldsymbol{F}$, we extract and transmit dynamic information along the time dimension through ConvGRU [55]. This process facilitates the interaction among peer-level features across different time steps, as illustrated by the vertical gray arrows in the HSF framework in Fig. 2.

In a ConvGRU cell, we take $\boldsymbol{D}_{t-1}^{l}$ from time $t-1$ on layer l as the hidden state, which also represents the previous dynamic feature. Through the convolutional transformations and gating

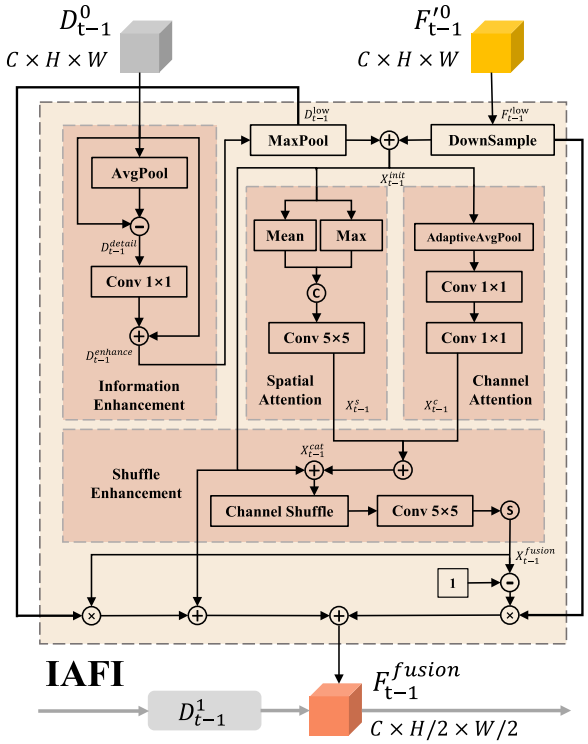


Fig. 4. Structure of IAFI. It consists of an information enhancement block, a shuffle enhancement block, and a multidimensional attention block.

mechanisms, we obtain the dynamic feature $\boldsymbol{D}_t^l$ at the current time. This design captures both spatial dependencies and temporal dynamics, making it effective for spatio-temporal fusion.

2) Interframe Asymmetric Feature Interaction: IAFI is designed to extract object dynamics more effectively and robustly, and the details are shown in Fig. 4. Shallow features primarily contain low-dimensional detail information, whose symmetric interaction requires a relatively stable interframe environment, and struggles to perceive dynamics under irregular camera jitter. In contrast, our asymmetric approach leverages the semantic stability of deep features across temporal scales, directing information flow from deep to shallow layers, which enhances the focus on moving object regions during shallow feature interactions.

At each time step, the static feature set $\boldsymbol{F}_t$ contains a series of scale-symmetric features $\{\boldsymbol{F}_t^0, \boldsymbol{F}_t^1, \ldots, \boldsymbol{F}_t^{\prime 1}, \boldsymbol{F}_t^{\prime 0}\}$. Here, we take the feature $\boldsymbol{F}_{t-1}^{\prime 0}$ and $\boldsymbol{D}_{t-1}^{0}$ as the input for IAFI. Prior to IAFI integration, $\boldsymbol{D}_{t-1}^{0}$ serves as ConvGRU's input with interaction limited to same-level interframe features, constraining its representation to low-level details. In contrast, the deep feature $\boldsymbol{F}_{t-1}^{\prime 0}$ from the preceding frame preserves both detailed features and high-level semantics. Based on this, IAFI enhances the representational capacity of $\boldsymbol{D}_{t-1}^{0}$ through $\boldsymbol{F}_{t-1}^{\prime 0}$'s feature guidance mechanism, enabling shallow motion information to fully benefit from the rich features in deep layers. And the output $\boldsymbol{F}_{t-1}^{\text{fusion}}$ of IAFI will serve as the input for ConvGRU, interacting with $\boldsymbol{D}_{t-1}^{1}$.

Specifically, to begin with, $\boldsymbol{D}_{t-1}^{0}$ strengthens its dynamic feature through an information enhancement block, resulting

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in $\boldsymbol{D}_{t-1}^{\text{enhance}}$. Following this, we downsample $\boldsymbol{D}_{t-1}^{\text{enhance}}$ and $\boldsymbol{F}_{t-1}^{\prime 0}$ to get $\boldsymbol{D}_{t-1}^{\text{low}}$ and $\boldsymbol{F}_{t-1}^{\prime \text{low}}$ which are then combined into $\boldsymbol{X}_{t-1}^{\text{init}}$. In the next step, we use an attention mechanism in both spatial and channel dimensions to help the model better capture key location information and enhance the focus on important feature channels. And based on the previous results, we can obtain $\boldsymbol{X}_{t-1}^{\text{cat}}$ for a channel shuffle operation [56], which enhances the information flow between channels by rearranging them across groups

$$\boldsymbol{X}_{t-1}^{\text{fusion}} = \sigma\left(\text{GC}_{5\times 5}\left(\text{CS}\left(\boldsymbol{X}_{t-1}^{\text{cat}}\right)\right)\right) \tag{1}$$

where σ is the sigmoid operation, CS$(\cdot)$ denotes the channel shuffle operation and GC denotes a group convolution layer. Finally, the result from IAFI can be obtained by

$$\begin{aligned}\boldsymbol{F}_{t-1}^{\text{fusion}} = \boldsymbol{X}_{t-1}^{\text{init}} + \boldsymbol{X}_{t-1}^{\text{fusion}} * \boldsymbol{D}_{t-1}^{\text{low}} \\ + \left(1 - \boldsymbol{X}_{t-1}^{\text{fusion}}\right) * \boldsymbol{F}_{t-1}^{\prime \text{low}}.\end{aligned} \tag{2}$$

D. Gaussian-Based Trajectory Association

In the trajectory association phase, most existing methods rely on either intersection over union (IoU)-based or feature embedding-based strategies. However, the tiny objects often lack distinctive appearance textures, rendering the matching schemes based on feature embeddings ineffective. In addition, the blurriness of small object boundaries leads to inaccurate distance metrics based on the bounding box. Furthermore, the uneven movements and irregular jittering of the satellite platform introduce substantial displacement between frames, posing challenges to the state update and prediction of the traditional Kalman filter.

To address these issues, we propose a trajectory association strategy based on Gaussian representation, which replaces bounding box-based object modeling (x, y, w, h) with an approach based on a 2-D-Gaussian distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. In this formulation, the center point of each box serves as the mean vector $\boldsymbol{\mu}$ of the Gaussian, while the square of half the side length defines the covariance matrix $\boldsymbol{\Sigma}$, namely,

$$\boldsymbol{\mu} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \left(\lambda \frac{w}{2}\right)^2 & 0 \\ 0 & \left(\lambda \frac{h}{2}\right)^2 \end{bmatrix} \tag{3}$$

where λ is used to adjust the spread degree of the Gaussian distribution. This approach eliminates the limitations of the uniform distribution by focusing on the probability distribution of the object-centric region, which better captures the characteristics of small objects. Moreover, this probabilistic representation better reflects the uncertainty of pixel displacement in a dual-motion coupled environment, effectively buffering observation errors. Building on this, inspired by Xu et al. [57], we employ the Kullback–Leibler divergence (KLD) to measure the distance between the objects and trajectories, which will be utilized as the cost for trajectory generation. The KLD between $\boldsymbol{B}_a$ and $\boldsymbol{B}_b$ is given as follows:

$$\begin{aligned}D_{\text{KL}}\left(B_a \parallel B_b\right) = \frac{1}{2}\Bigg(\left(\mu_b - \mu_a\right)^{\top} \Sigma_b^{-1}\left(\mu_b - \mu_a\right) \\ + tr\left(\Sigma_b^{-1}\Sigma_a\right) + \ln\frac{|\Sigma_b|}{|\Sigma_a|} - 2\Bigg).\end{aligned} \tag{4}$$

By introducing the KLD-based metric, we achieve robust trajectory association under motion interference conditions and provide relatively stable observations for the Kalman filter. Furthermore, to mitigate erroneous corrections caused by anomalous deviations between unstable measurements and the prior distribution, a KLD-based regularization term is introduced during state updates in the Kalman filter. The optimized Kalman gain is defined as follows:

$$\tilde{K} = K \cdot \frac{\epsilon}{\epsilon + D_{\text{KL}}\left(\mathcal{N}\left(\hat{x}, P\right) \| \mathcal{N}\left(\hat{x}^{-}, P^{-}\right)\right)} \tag{5}$$

where K is the common Kalman gain and ϵ is the KLD threshold between the prior and posterior distributions.

E. Loss Function

To ensure effective training of the model, we employ focal loss [27] and $L1$ loss for multitask supervision. Specifically, considering the extreme imbalance between foreground and background in remote sensing scenarios, the focal loss is employed in the design of our loss function

$$\mathcal{L}_{\text{focal}} = -\frac{1}{N}\sum_{i=1}^{N}\begin{cases}(1-p_i)^{\gamma}\log\left(p_i\right)x, & \text{if } y_i = 1 \\ (1-y_i)^{\beta}\, p_i^{\gamma}\log\left(1-p_i\right), & \text{otherwise}\end{cases} \tag{6}$$

where N represents the number of the samples, while i represents a single sample, $y_i = 1$ corresponds to positive points, and $p_i \in [0, 1]$ corresponds to the predicted probability. β and γ are hyper parameters. In addition, $L1$ loss is applied to distance measurement

$$\mathcal{L}_{L1} = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \hat{y}_i\right| \tag{7}$$

where y_i denotes the ground-truth value of the ith sample, $\hat{y}_i$ denotes the predicted value of the ith sample, and N represents the total number of samples.

The overall loss function $\mathcal{L}$ of our method is designed to supervise multiple tasks, including heatmap prediction, bounding box size regression, offset regression, and sequence modeling. It is formulated as a weighted sum of four components

$$\mathcal{L} = \lambda_{\text{hm}}\mathcal{L}_{\text{hm}} + \lambda_{\text{wh}}\mathcal{L}_{\text{wh}} + \lambda_{\text{off}}\mathcal{L}_{\text{off}} + \lambda_{\text{seq}}\mathcal{L}_{\text{seq}} \tag{8}$$

where λ_{hm}, λ_{wh}, λ_{off}, and λ_{seq} are weighting factors for the heatmap, box size, offset, and sequence losses, respectively. $\mathcal{L}_{\text{hm}}$ supervises the center-point heatmap using the focal loss

$$\mathcal{L}_{\text{hm}} = \frac{1}{k}\sum_{t=1}^{k}\mathcal{L}_{\text{focal}}\left(\hat{\mathbf{H}}_t, \mathbf{H}_t\right) \tag{9}$$

where $\hat{\mathbf{H}}_t$ and $\mathbf{H}_t$ are the predicted and Gaussian label map distributed at the center of the positive samples at time step t. k denotes the length of the input sequence. $\mathcal{L}_{\text{wh}}$ penalizes errors in the predicted object sizes using a regression loss

$$\mathcal{L}_{\text{wh}} = \frac{1}{k}\sum_{t=1}^{k}\mathcal{L}_{L1}\left(\hat{\mathbf{S}}_t, \mathbf{S}_t\right) \tag{10}$$

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where $\hat{\mathbf{S}}_t$ and $\mathbf{S}_t$ denote the predicted and ground truth object size. $\mathcal{L}_{\text{off}}$ measures the difference between predicted and ground truth center offsets

$$\mathcal{L}_{\text{off}} = \frac{1}{k}\sum_{t=1}^{k}\mathcal{L}_{L1}\left(\hat{\mathbf{O}}_t, \mathbf{O}_t\right) \tag{11}$$

where $\hat{\mathbf{O}}_t$ and $\mathbf{O}_t$ are the predicted and ground truth offsets. $\mathcal{L}_{\text{seq}}$ supervises the predicted masks which are used for consistency constraint

$$\mathcal{L}_{\text{seq}} = \frac{1}{k}\sum_{t=1}^{k}\mathcal{L}_{\text{focal}}\left(\hat{\mathbf{M}}_t^{\text{res}}, \mathbf{M}_t^{\text{res}}\right) \tag{12}$$

where $\hat{\mathbf{M}}_t^{\text{res}}$ and $\mathbf{M}_t^{\text{res}}$ are the predicted and ground truth masks.

IV. Experiments

A. Datasets and Metrics

We use SatVideoDT [10] and SatMTB-MOT [6], two widely used datasets in the satellite video moving object tracking field, to evaluate our method. SatVideoDT is publicly available through the SatVideoDT Challenge[1]: and includes 114 satellite videos, with 82 for training, 11 for validation, and 21 for testing. This dataset only contains the vehicle category and comprehensively covers various small object scenarios. SatMTB-MOT is the latest dataset dedicated to multiclass moving object tracking in satellite videos, covering a broader range of application scenarios. Among the 224 videos, 62 focus on airplanes, 70 on ships, and 92 on vehicles. We analyze the model's performance in multiclass scenarios based on this dataset. Regarding evaluation metrics, we use the MOT metrics [58] to analyze tracking performance quantitatively. The MOT metrics include multiple object tracking accuracy (MOTA), false positive (FP), false negative (FN), precision (Prcn), recall (Rcll), identity switches (IDs), the percentage of mostly tracked (MT), and mostly lost (ML) trajectories and identity F1 score (IDF1). Among these metrics, MOTA highlights tracking precision and detection completeness, IDF1 emphasizes temporal identity consistency, and both are complementary for a holistic measure of tracking quality. In addition, parameters (Params), floating point operations (FLOPs), and frames/s (FPS) are adopted to measure the efficiency of the models. For a fair comparison, the efficiency of TBD-based methods is measured by considering both the detection and tracking stages.

B. Implementation Details

We use DLA-34 [54] as the encoder and employ a decoder head similar to that in CenterTrack [16]. Based on this unified framework, we develop two versions of our method, SPAX and SPAX-Light. Both share the same overall architecture, while SPAX-Light reduces the number of layers in the HSF module to further improve efficiency. We apply the Adam optimizer, with initial learning rates of 1.25×10^{-4} and 1×10^{-4} for the SatVideoDT [10] and SatMTB-MOT [6] dataset, respectively. We train on the SatVideoDT dataset for 20 epochs, with learning rate decay applied at the 18th epoch, and train on the SatMTB-MOT dataset for 30 epochs, with decay applied at the 28th epoch. During the training phase, we crop the input images to a size of 512×512 pixels and perform simple data augmentations, such as random mirroring and flipping. Our method maintains the original image size and excludes all forms of test-time augmentation during the inference phase. The consecutive frame number m for motion priors is set to 20, the frame number k for network input is set to 5 for SPAX and 3 for SPAX-Light, mask threshold α is set to 5, consistency threshold s is set to 0.2 and the λ in GTA is set to 3. About region radius r, we use a range of values, where smaller values result in higher model inference efficiency but lower accuracy, and vice versa. We implement our sparse network in Python using the PyTorch framework [59] and write custom CUDA kernels based on previous methods [21]. It is worth noting that the model's fully sparse computation is only conducted during the inference stage and this part is training-free and plug-and-play. The first frame of each video still uses a dense computation mode. Our model is trained on two RTX 3090 GPUs and tested on one Titan Xp GPU.

C. Comparison to the State-of-the-Arts

In the experimental section, we compare our method with the SOTA trackers, which include generic trackers such as DeepSORT [30], Tracktor [31], CenterTrack [16], FairMOT [18], ByteTrack [17], as well as satellite trackers CFTracker [14] CKDNet-SMTNet [11], DSFNet [12], Hieum [13], and MP^{2}Net [15]. For CenterTrack, we adjust it with a low output feature map resolution (stride 2) to accommodate small objects. For DSFNet and Hieum, we apply SORT [29] as the tracker to generate the final results, which is consisted with the open-source code. Notably, Hieum is originally designed for offline moving object detection and we adjust its inference process to suit the online setting. In addition, considering the postprocessing part in MP^{2}Net uses results from future frames, we modify it to strictly align with our setting for a fair comparison.

We conduct experiments on the SatVideoDT dataset [10] and the results are shown in Table I. Our SPAX achieves the best performance among all the trackers. Compared to general trackers, our method achieves a multiple-fold improvement across several metrics. For example, our MOTA is 2.3 times that of ByteTrack [17] (66.2% versus 28.2%), and our MT is 3 times higher (869 versus 288). The absolute dominance in both overall tracking accuracy and long-term success rate underscores the strongly temporal dependency inherent in this task, while simultaneously demonstrating the exceptional capability of HSF in effectively mining and utilizing temporal information. For the trackers in remote sensing, SPAX outperforms the best baseline [15] by 5.1% MOTA and 4.5% IDF1. In addition, the increase in MT (from 710 to 869) and the decrease in ML (from 207 to 135) demonstrate our method's ability to successfully track more objects. It is worth noting that the IDs are only 275, which indicates that we have achieved more robust tracking. Furthermore, our SPAX-Light maintains leading performance in most accuracy metrics, achieving an excellent balance between accuracy and

[1]https://satvideodt.github.io

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TABLE I

QUANTITATIVE RESULTS ON THE SATVIDEODT DATASET [10]. THE GENERIC TRACKERS AND SATELLITE-SPECIFIC TRACKERS ARE POSITIONED ABOVE AND BELOW THE DIVIDING LINE, RESPECTIVELY. THE BEST RESULTS AMONG THE SATELLITE-SPECIFIC TRACKERS ARE HIGHLIGHTED IN RED AND THE SECOND-BEST IN GREEN

Method	Type	**MOTA** ↑	**IDF1** ↑	Rcll ↑	Prcn ↑	FP ↓	FN ↓	MT ↑	ML ↓	IDs ↓	Params (M) ↓	FLOPs(G) ↓	FPS ↑
DeepSort [30]	TBD	14.4%	32.2%	27.4%	71.0%	31448	204177	170	706	5048	86.2	260.8	3.3
Tracktor [31]	JDT	6.0%	18.2%	17.2%	63.2%	28094	232901	43	927	3428	41.3	210.5	2.1
Fairmot [18]	JDT	11.2%	21.3%	17.9%	82.9%	10399	230767	86	890	8423	16.6	231.2	8.6
ByteTrack [17]	TBD	28.2%	45.6%	33.2%	87.1%	13782	187730	288	698	281	99.0	361.0	8.5
CenterTrack [16]	JDT	38.3%	50.1%	57.5%	75.8%	51488	119617	601	311	2245	1.9	147.7	5.2
CFTracker [14]	JDT	35.4%	49.0%	59.3%	72.0%	64896	114434	621	292	2346	2.0	147.9	5.0
CKDNet-SMTNet [11]	TBD	39.6%	57.7%	50.2%	83.1%	28759	140092	324	301	934	78.1	1700.5	0.8
DSFNet [12]	TBD	52.3%	66.1%	56.2%	93.8%	10418	123231	525	291	543	6.7	216.7	1.6
Hieum [13]	TBD	59.5%	71.0%	62.6%	95.6%	8089	105241	619	207	490	1.2	123.7	3.5
MP^{2}Net [15]	JDT	61.1%	73.1%	65.8%	93.6%	12721	96241	710	207	403	6.7	774.1	1.1
SPAX (Ours)	TBD	66.2%	77.6%	74.7%	89.9%	23563	71198	869	135	275	7.9	413.9	1.9
SPAX-Light (Ours)	TBD	58.8%	71.8%	66.3%	90.0%	20588	94845	759	206	350	7.1	42.3	7.3

TABLE II

QUANTITATIVE RESULTS ON THE SATMTB-MOT (VEHICLE) DATASET [6]. THE GENERIC TRACKERS AND SATELLITE-SPECIFIC TRACKERS ARE POSITIONED ABOVE AND BELOW THE DIVIDING LINE, RESPECTIVELY. THE BEST RESULTS AMONG THE SATELLITE-SPECIFIC TRACKERS ARE HIGHLIGHTED IN RED AND THE SECOND-BEST IN GREEN

Method	Type	**MOTA** ↑	**IDF1** ↑	Rcll ↑	Prcn ↑	FP ↓	FN ↓	MT ↑	ML ↓	IDs ↓	Params (M) ↓	FLOPs(G) ↓	FPS ↑
DeepSort [30]	TBD	11.2%	33.1%	27.9%	65.4%	50633	248045	335	1900	6822	86.2	260.9	4.6
Tracktor [31]	JDT	15.9%	41.5%	34.9%	65.4%	63596	223843	500	1570	1626	41.3	210.5	3.4
CenterTrack [16]	JDT	35.7%	45.2%	38.3%	95.3%	6479	212170	760	1580	2438	1.8	147.7	5.5
CFTracker [14]	JDT	48.0%	60.3%	63.4%	81.1%	50854	125719	1546	751	2325	1.9	147.8	5.1
CKDNet-SMTNet [11]	TBD	25.6%	46.7%	37.9%	76.2%	40776	213647	242	1136	1352	78.0	1260.6	2.9
DSFNet [12]	TBD	55.9%	68.9%	59.6%	94.6%	11817	138889	1295	790	939	6.7	60.6	4.4
Hieum [13]	TBD	56.1%	70.7%	74.0%	81.0%	59575	89277	1976	407	2198	1.2	17.7	7.7
MP^{2}Net [15]	JDT	54.3%	68.0%	57.1%	95.6%	9016	147602	1290	949	659	8.1	216.9	3.7
SPAX (Ours)	TBD	58.4%	71.9%	64.2%	91.9%	19500	123020	1548	747	524	9.2	466.1	4.3
SPAX-Light (Ours)	TBD	54.3%	68.3%	58.1%	94.1%	12485	144131	1323	909	646	8.4	57.0	15.4

efficiency. Specifically, it achieves comparable accuracy to Hieum (−0.7 MOTA and +0.8 IDF1) while requiring only 34% of its computational cost and delivering a **2.1**× increase in FPS.

In addition, our method achieves the best performance in terms of recall and FN, while exhibiting slightly lower precision and higher FP. This reflects the typical "recall–precision tradeoff" commonly observed in detection–tracking systems, which is usually influenced by the choice of detection and tracking thresholds. However, we do not perform dataset-specific fine-tuning of these thresholds. Instead, the same configuration is applied across all datasets. Even under this unified setting, our SPAX attains a recall nearly 10 percentage points higher than the best existing approaches while maintaining a precision of 90%, clearly demonstrating its overall superiority.

Fig. 5 depicts some visualized results on SatVideoDT-002 which is an urban dense environment equipped with narrow streets, small objects, and interference from various city facilities, posing considerable challenges. CenterTrack exhibits clear disadvantages in the moving object tracking task. It struggles to distinguish between foreground and background in many cases, resulting in a significant number of false detections. In addition, its trajectory association method, which relies on predicting offsets, shows poor robustness in a dual-motion system, making long-term and stable tracking difficult. This leads to numerous fragmented trajectories, with IDs reaching 290. Although DSFNet and MP^{2}Net surpass previous methods and achieve great accuracy, they still suffer from a large number of false positives and false negatives in complex scenarios, with some localization errors in the detected objects. Furthermore, both methods exhibit instability during long-term tracking, leading to more frequent trajectory switches. Benefiting from our comprehensive spatio-temporal fusion strategy and Gaussian-based object modeling, our method demonstrates superior performance in both short-term (100th frame) and long-term (300th frame) tracking scenarios. It achieves the highest true positive (TP) rate while maintaining the lowest FP and FN rates, reaching long-term and stable tracking.

Experimental results (see Table II) on SatMTB-MOT-vehicle [6] validation set demonstrate the performance advantages of our method across key metrics. For SPAX, the MOTA and IDF1 achieve 58.4% and 71.9%, respectively, which indicates superior performance in tracking accuracy and identity consistency, allowing for more reliable tracking in complex scenarios. Besides, our approach strikes a strong balance between Prcn (91.9%) and Rcll (64.2%), maintaining a low number of IDs (524), which showcases its robust tracking capability. In addition, SPAX-Light attains performance on par with the baseline MP^{2}Net, yet its computational complexity is reduced to 26%, with a remarkable **4.2**× improvement in inference speed (achieving 15.4 FPS), which fully satisfies the demands of real-world applications.

The SatMTB-MOT-vehicle dataset is characterized by relatively low image quality and strong external interference. From the results, Hieum [13] achieves highly competitive results compared with our model. However, Hieum employs a multiframe feature interaction window of length 20, which is more effective in mitigating short-term noise fluctuations [15]

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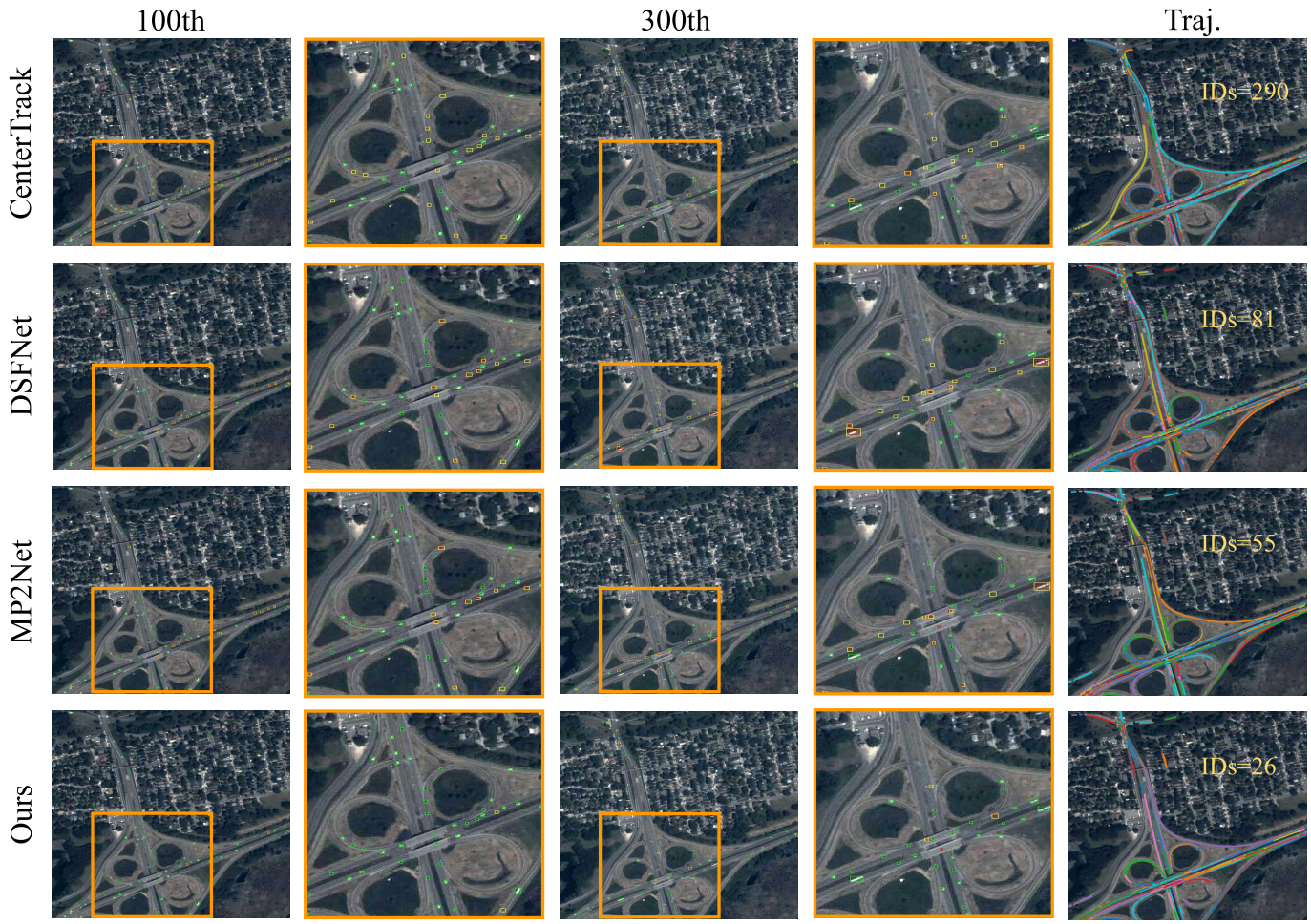


Fig. 5. Visual results of CenterTrack, DSFNet, MP^{2}Net, and SPAX. The first and third columns represent the results of the 100th and 300th frames of the SatVideoDT-002 sequence [10] followed by the corresponding locally enlarged views, respectively. The TP corresponds to the green box, the FN corresponds to the yellow box, and the FP corresponds to the red box. The last image in each row shows the tracked trajectories of the whole sequence. Different colors represent different trajectories.

compared to our historical feature interaction module with a shorter window length of 5. Even under such challenging conditions, our method achieves comparable or even superior performance, and using only one-fourth of Hieum's temporal window length not only reduces memory access overhead but also enables more efficient throughput. Furthermore, it is noteworthy that on the subsequent SatMTB-MOT-ship and SatMTB-MOT-airplane datasets [6], our model with a window length of 5 significantly outperforms Hieum with a window length of 20, further demonstrating the superiority and generalization capability of our approach.

Fig. 6 provides a set of visualizations on a low signal-to-noise ratio (SNR) and densely populated scene. This SatMTB-MOT-vehicle-23 sequence exhibits low clarity and the objects are blurry and difficult to distinguish. Even under such challenging conditions, our method demonstrates stable and reliable performance. Specifically, it achieves a relatively higher recall rate and maintains a high level of accuracy with few false positives, even when the image quality is poor.

In the following, we test the model's performance on multiclass objects using SatMTB-MOT-ship and SatMTB-MOT-airplane datasets [6], as shown in Tables III and IV. On SatMTB-MOT-ship validation set, SPAX ranks 1st among all the trackers. It outperforms the second-performance tracker [15] by a large margin (i.e., +7.6 MOTA and +4.6 IDF1). Meanwhile, our method achieves relatively high MT (44) and the lowest ML (20). It is noteworthy that while some methods [12], [13], [15] demonstrate satisfactory performance on the vehicle category in the SatVideoDT dataset, they exhibit significant limitations when applied to the ship category in the SatMTB-MOT dataset. In contrast, our approach maintains consistently superior performance across diverse image domains and various object categories, achieving superior cross-domain adaptability and effective category generalization. Furthermore, our SPAX-Light achieves comprehensive superiority in both accuracy and speed: its MOTA surpasses that of the previous SOTA method MP^{2}Net by 6.6 points, while reducing computation by 84% and boosting inference speed by **5.9**×.

For the SatMTB-MOT-airplane validation set, our method also exhibits notable improvements across several key metrics, including MOTA, IDF1, MT, ML, and Rcll. Specifically, SPAX achieves 76.1% on MOTA, outperforming MP^{2}Net (72.7%) and Hieum (53.9%). In terms of IDF1, our method

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TABLE III

QUANTITATIVE RESULTS ON THE SATMTB-MOT (SHIP) DATASET [6]. THE GENERIC TRACKERS AND SATELLITE-SPECIFIC TRACKERS ARE POSITIONED ABOVE AND BELOW THE DIVIDING LINE, RESPECTIVELY. THE BEST RESULTS AMONG THE SATELLITE-SPECIFIC TRACKERS ARE HIGHLIGHTED IN RED AND THE SECOND-BEST IN GREEN

Method	Type	**MOTA** ↑	**IDF1** ↑	Rcll ↑	Prcn ↑	FP ↓	FN ↓	MT ↑	ML ↓	IDs ↓	Params (M) ↓	FLOPs(G) ↓	FPS ↑
DeepSort [30]	TBD	41.8%	65.7%	63.5%	74.7%	4687	7391	45	17	40	86.2	239.0	10.5
Tracktor [31]	JDT	30.5%	55.3%	63.9%	66.6%	6975	7855	38	15	280	41.3	231.1	3.5
CenterTrack [16]	JDT	42.1%	63.6%	55.8%	80.4%	2959	9599	36	22	20	1.8	147.7	5.3
CFTracker [14]	JDT	29.6%	63.9%	63.9%	65.1%	7436	7856	48	22	5	1.9	147.8	5.2
CKDNet-SMTNet [11]	TBD	17.1%	39.3%	29.5%	70.8%	2647	15335	15	38	42	78.0	1576.5	1.4
DSFNet [12]	TBD	41.5%	60.3%	53.6%	82.2%	2516	10089	28	27	108	6.7	157.8	2.2
Hieum [13]	TBD	24.3%	47.2%	37.0%	74.7%	2725	13706	22	40	24	1.2	47.4	5.8
MP^{2}Net [15]	JDT	44.6%	66.0%	60.3%	79.6%	3354	8631	37	20	60	8.1	552.4	1.5
SPAX (Ours)	TBD	52.2%	70.6%	62.8%	85.6%	2290	8081	44	20	13	9.2	186.2	3.8
SPAX-Light (Ours)	TBD	51.2%	70.3%	60.8%	86.4%	2072	8521	44	24	20	8.4	90.5	8.8

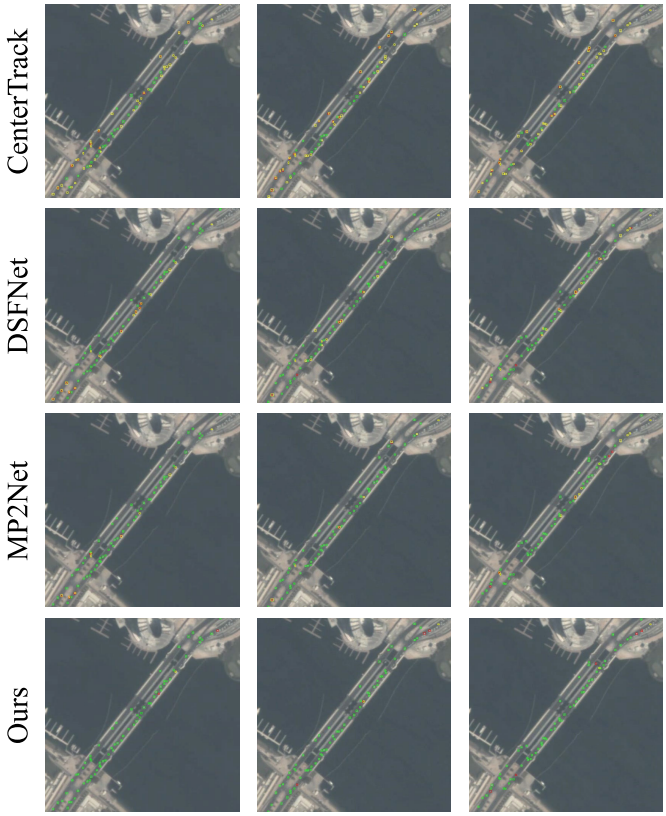


Fig. 6. Visual results of CenterTrack, DSFNet, MP^{2}Net, and SPAX. The three columns represent the results of the 30th, 60th, and 90th frames of the SatMTB-MOT-Vehicle-23 sequence [6], respectively. The TP corresponds to the green box, the FN corresponds to the yellow box, and the FP corresponds to the red box.

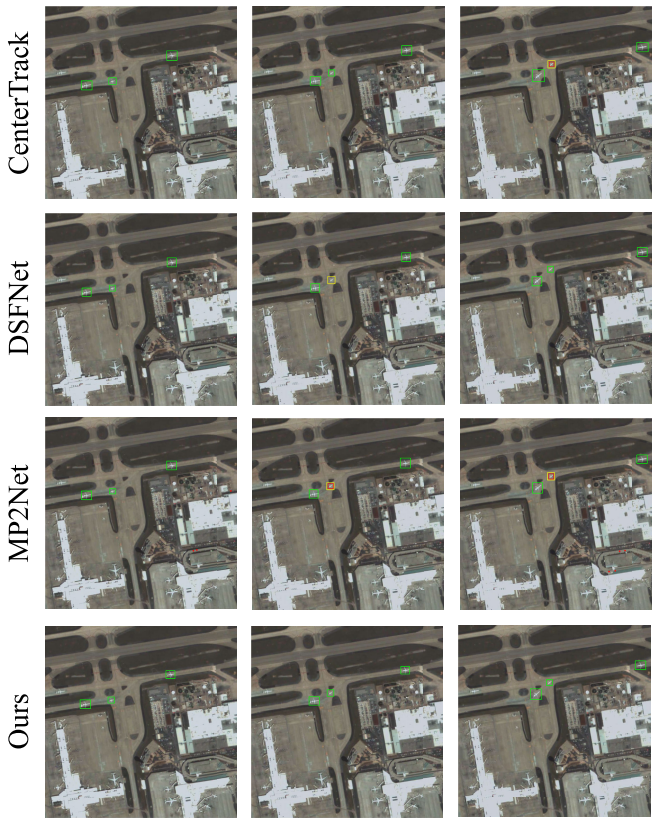


Fig. 7. Visual results of CenterTrack, DSFNet, MP^{2}Net, and SPAX. The three columns represent the results of the 50th, 150th, and 250th frames of the SatMTB-MOT-Airplane-5 sequence [6], respectively. The TP corresponds to the green box, the FN corresponds to the yellow box, and the FP corresponds to the red box.

achieves an impressive score of 86.7%, surpassing MP^{2}Net (83.2%) and all other competitors, highlighting its exceptional capability in maintaining consistent object identities across frames. Moreover, our Rcll (82.3%) is marginally higher than MP^{2}Net (75.9%), reflecting strong performance in detecting and tracking the large-sized objects. The results, reflected by an MT of 51 and a low ML score of 5, further underscore the robustness of our approach in tracking objects over time while minimizing object losses. In addition, our SPAX-Light attains a **5.3**× acceleration over MP^{2}Net with negligible accuracy degradation. Overall, these results establish our method as a highly effective solution for moving object tracking, capable of handling the challenges posed by multiscale objects and demonstrating excellent adaptability across multiple categories.

Figs. 7 and 8 present the visual results for SatMTB-MOT-Airplane-5 sequence [6] and SatMTB-MOT-Ship-38 sequence [6]. We compare our method with CenterTrack [16], DSFNet [12], and MP^{2}Net [15]. The airplane and ship categories, compared to vehicles, have larger scales and more distinct features, which reduces the tracking difficulties. Even so, in a long-term tracking situation, existing algorithms still struggle to ensure accurate and stable perception. In contrast, by leveraging HSF and GTA, our method achieves precise and

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TABLE IV

QUANTITATIVE RESULTS ON THE SATMTB-MOT (AIRPLANE) DATASET [6]. THE GENERIC TRACKERS AND SATELLITE-SPECIFIC TRACKERS ARE POSITIONED ABOVE AND BELOW THE DIVIDING LINE, RESPECTIVELY. THE BEST RESULTS AMONG THE SATELLITE-SPECIFIC TRACKERS ARE HIGHLIGHTED IN RED AND THE SECOND-BEST IN GREEN

Method	Type	**MOTA** ↑	**IDF1** ↑	Rcll ↑	Prcn ↑	FP ↓	FN ↓	MT ↑	ML ↓	IDs ↓	Params (M) ↓	FLOPs(G) ↓	FPS ↑
DeepSort [30]	TBD	51.9%	74.2%	70.1%	79.4%	2837	4676	43	13	3	86.2	221.7	10.6
Tracktor [31]	JDT	17.2%	57.0%	60.3%	58.4%	6701	6204	31	19	24	41.3	227.7	3.6
CenterTrack [16]	JDT	49.8%	71.4%	63.1%	82.6%	2075	5770	40	21	2	1.8	147.7	5.1
CFTracker [14]	JDT	44.1%	71.7%	70.7%	72.7%	4152	4580	45	16	5	1.9	147.8	4.7
CKDNet-SMTNet [11]	TBD	50.8%	66.4%	52.6%	96.8%	271	7396	33	17	14	78.0	1737.2	1.0
DSFNet [12]	TBD	63.3%	77.2%	66.9%	95.1%	544	5166	40	9	17	6.7	219.5	1.6
Hieum [13]	TBD	53.9%	71.0%	59.3%	92.0%	808	6351	36	12	34	1.2	68.7	4.8
MP^{2}Net [15]	JDT	72.7%	83.2%	75.9%	96.1%	484	3765	48	6	18	8.1	628.1	1.0
SPAX (Ours)	TBD	76.1%	86.7%	82.3%	93.1%	953	2766	51	5	8	9.2	871.3	1.6
SPAX-Light (Ours)	TBD	72.3%	81.5%	77.0%	94.4%	714	3588	46	5	29	8.4	207.8	5.3

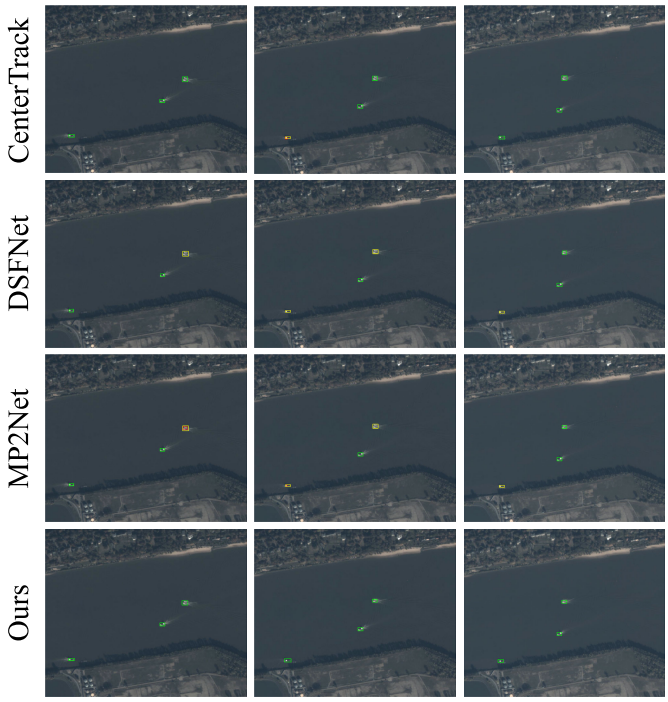


Fig. 8. Visual results of CenterTrack, DSFNet, MP^{2}Net, and SPAX. The three columns represent the results of the 50th, 150th, and 250th frames of the SatMTB-MOT-Ship-38 sequence [6], respectively. The TP corresponds to the green box, the FN corresponds to the yellow box, and the FP corresponds to the red box.

robust tracking, demonstrating its strong advantage in handling multiple categories in different scenarios.

Moreover, we select four typical large scenarios in SatMTB-MOT dataset [6] where airplanes, ships, and vehicles coexist and visualize the performance of our tracker in Fig. 9. In this case, our tracker can simultaneously detect and track the moving large airplanes, smaller ships, and tiny vehicles, which proves the ability to handle multiscale objects of our model.

D. Ablation Studies

In this section, we conduct ablation experiments to investigate the effectiveness of the fully sparse paradigm, IAFI, GTA, and foreground mask generation. Note that all the ablation experiments are conducted based on SPAX.

TABLE V

COMPARISON OF THE ACCURACY AND EFFICIENCY OF OUR METHOD WITH THE BASELINE UNDER DIFFERENT SETTINGS ON SATVIDEODT [10] AND SATMTB-MOT DATASET [6]

Dataset	Category	Method	Radius	MOTA ↑	IDF1 ↑	FPS ↑	Speedup
SatVideoDT	Vehicle	MP^{2}Net	-	61.1%	73.1%	1.1	1.0×
		Ours	-	65.9%	77.3%	1.7	1.5×
			20	62.4%	74.8%	3.0	2.7×
			15	60.4%	73.5%	3.1	2.8×
			10	55.2%	69.9%	3.2	2.9×
SatMTB-MOT	Vehicle	MP^{2}Net	-	54.3%	68.0%	3.7	1.0×
		Ours	-	58.4%	71.9%	4.3	1.2×
			20	55.9%	69.1%	7.0	1.9×
			15	53.5%	67.4%	7.5	2.0×
			10	49.8%	64.3%	7.8	2.1×
	Airplane	MP^{2}Net	-	72.7%	83.2%	1.0	1.0×
		Ours	-	76.2%	86.7%	1.4	1.4×
			60	72.2%	83.2%	3.0	3.0×
			50	70.2%	81.8%	3.2	3.2×
			40	69.4%	81.6%	3.3	3.3×
	Ship	MP^{2}Net	-	44.6%	66.0%	1.5	1.0×
		Ours	-	53.3%	72.6%	1.9	1.3×
			30	50.0%	68.6%	4.5	3.0×
			25	47.7%	66.7%	4.6	3.1×
			20	40.8%	58.3%	4.7	3.1×

1) Effectiveness of the Fully Sparse Paradigm: In this section, we provide a visual comparison of the feature maps in Fig. 10 for intuitive understanding. Furthermore, we present the comparative results for multiple categories across two datasets with the baseline model MP^{2}Net [15] in Table V. Extensive experiments show that the fully sparse framework achieves multifold acceleration compared to the baseline while maintaining high accuracy. Specifically, for the ship category in the SatMTB-MOT dataset, our fully sparse model achieves 3.1× speedup while reaching SOTA accuracy by exploiting the fact that ships are generally sparsely distributed over expansive maritime regions and occupy only a limited spatial extent in high-resolution imagery. This demonstrates that our method effectively addresses the issue of background redundancy in the spatial dimension in remote sensing scenes, resulting in significant efficiency gains.

In addition, we compare the parameters (Params) and FLOPs between our model and the baseline model [15] in Table VI. For SatVideoDT dataset [10], although our model has a slightly higher Params (7.9 M) compared to the baseline (6.7 M) derived from temporal enhancements, it significantly reduces FLOPs from 774.1 to 89.5 G, achieving an 8.6× compression ratio (an 88.4% reduction) through spatially sparse computation. SatMTB-MOT (ship) dataset [6] exhibits consistent outcomes, and it is noteworthy that our method delivers a 14.7× compression rate (a 93.2% reduction)

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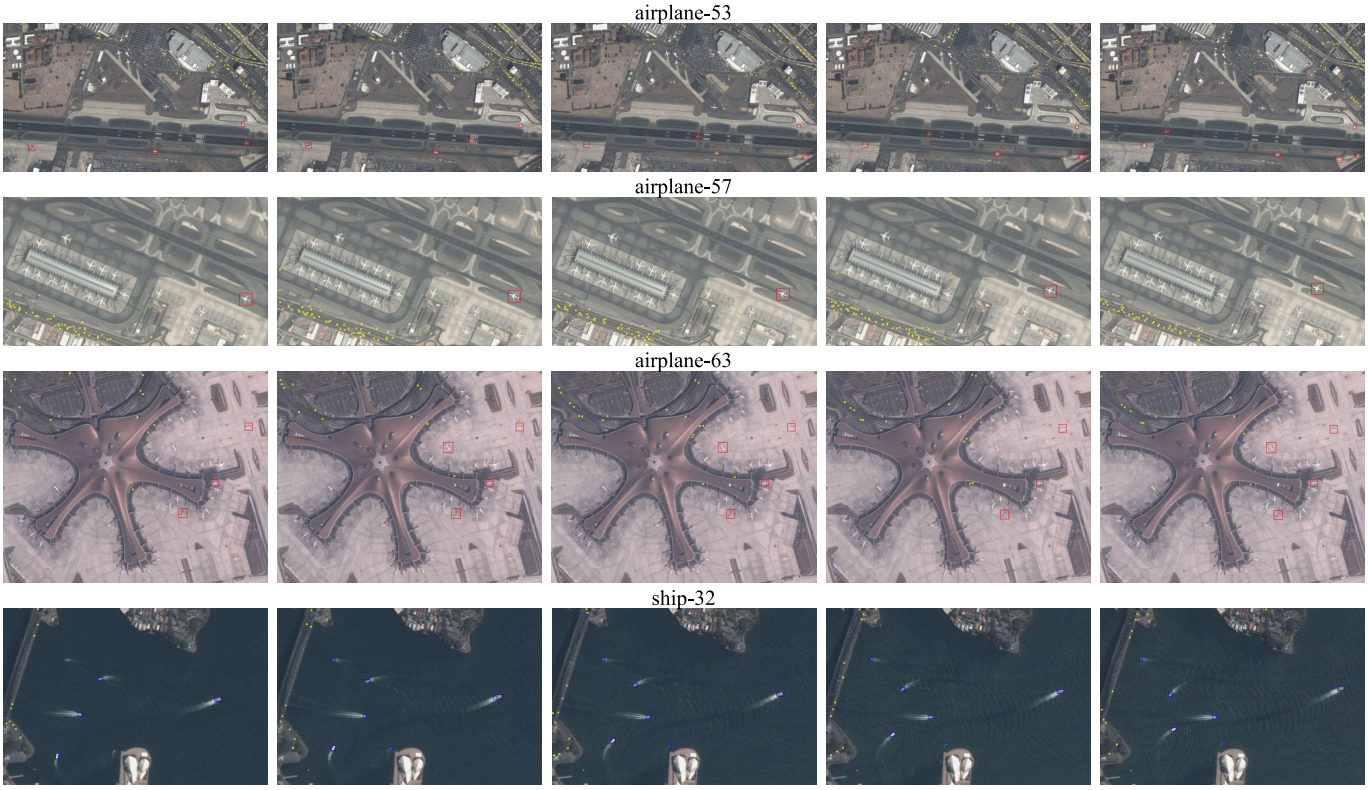


Fig. 9. Multicategory tracking performance of our SPAX on the SatMTB-MOT dataset [6]. Each row represents an independent sequence. The yellow box denotes the tracked vehicle, the red box denotes the airplane, and the blue box denotes the ship. Zoomed-in view to view better.

TABLE VI

COMPREHENSIVE COMPARISON BETWEEN OUR MODEL AND THE BASELINE MODEL ON THE SATVIDEODT DATASET [10] AND SATMTB-MOT (SHIP) DATASET [6]. THE RESULTS ARE REPRODUCED BY US

Dataset	Method	MOTA ↑	Params (M) ↓	FLOPs (G) ↓		FPS ↑	
				Statistics	Compression Ratio	Statistics	Speedup
SatVideoDT	MP^{2}Net	61.1%	6.7	774.1	1×	1.1	1×
	Ours	62.4%	7.9	89.5	**8.6**×	3.0	**2.7**×
SatMTB-MOT	MP^{2}Net	44.6%	8.1	552.4	1×	1.5	1×
	Ours	45.8%	9.2	37.6	**14.7**×	4.6	**3.1**×

TABLE VII

ABLATION STUDIES OF THE IAFI AND GTA IN SPAX

Baseline	IAFI	GTA	MOTA ↑	IDF1 ↑	MT ↑	ML ↓	IDs ↓
✓			61.1%	73.1%	710	207	403
✓	✓		64.5%	75.2%	765	155	405
✓	✓	✓	66.2%	77.6%	869	135	275

TABLE VIII

RESULTS OF APPLYING GTA TO FIVE DIFFERENT TRACKERS ON THE SATVIDEODT VALIDATION SET [10]

Method	Similarity	GTA	MOTA (%) ↑	IDF1 (%) ↑	IDs ↓
ByteTrack [17]	Motion (Byte)		28.2	45.6	281
	Motion (Byte)	✓	31.0 +2.8	47.9 +2.3	277 +1.4%
DSFNet [12]	Motion (Sort)		52.3	66.1	543
	Motion (Sort)	✓	52.4 +0.1	66.7 +0.6	503 +7.4%
DeepSort [30]	Motion + ReID		14.4	32.2	5048
	Motion + ReID	✓	14.2 -0.2	32.9 +0.7	3459 +31.5%
Fairmot [18]	Motion + ReID		11.2	21.3	8423
	Motion + ReID	✓	16.9 +5.7	28.3 +7.0	1478 +82.5%
	Motion	✓	17.0 +5.8	28.6 +7.3	1373 +83.7%
CenterTrack [16]	Motion		38.3	50.1	2245
	Motion	✓	42.3 +4.0	64.7 +14.6	629 +72.0%

relative to the baseline. These results validate the mismatch in computational resource allocation: redundant spatial feature computation and insufficient temporal utilization. By employing a more comprehensive temporal strategy and eliminating redundant background feature computations, our SPAX provides a more efficient overall solution.

2) Interframe Asymmetric Feature Interaction: IAFI is designed to extract object dynamics more effectively and robustly. The ablation study on the SatVideoDT dataset [10] shows a clear progression in tracking performance as IAFI added, as shown in Table VII. Starting with the baseline [15], the introduction of IAFI improves MOTA from 61.1% to 64.5%, IDF1 from 73.1% to 75.2%, and MT from 710 to 765, while reducing ML from 207 to 155. Fig. 11 further shows the effectiveness of IAFI. As can be seen from the feature maps of applying IAFI, our method demonstrates remarkable performance in detail preservation and object saliency enhancement, enabling a more clear delineation of small moving objects and boundary regions, while significantly improving the distinguish ability between dynamic foreground and unstable background. Moreover, IAFI shows distinct advantages in noise suppression and feature robustness, illustrating smooth low-response regions even if in densely distributed scenes. All these clearly indicate the substantial gains brought by deep rich features to the extraction of shallow motion information, demonstrating the effectiveness of IAFI.

3) Gaussian-Based Trajectory Association: GTA overcomes the uniform distribution limitations in dual-motion

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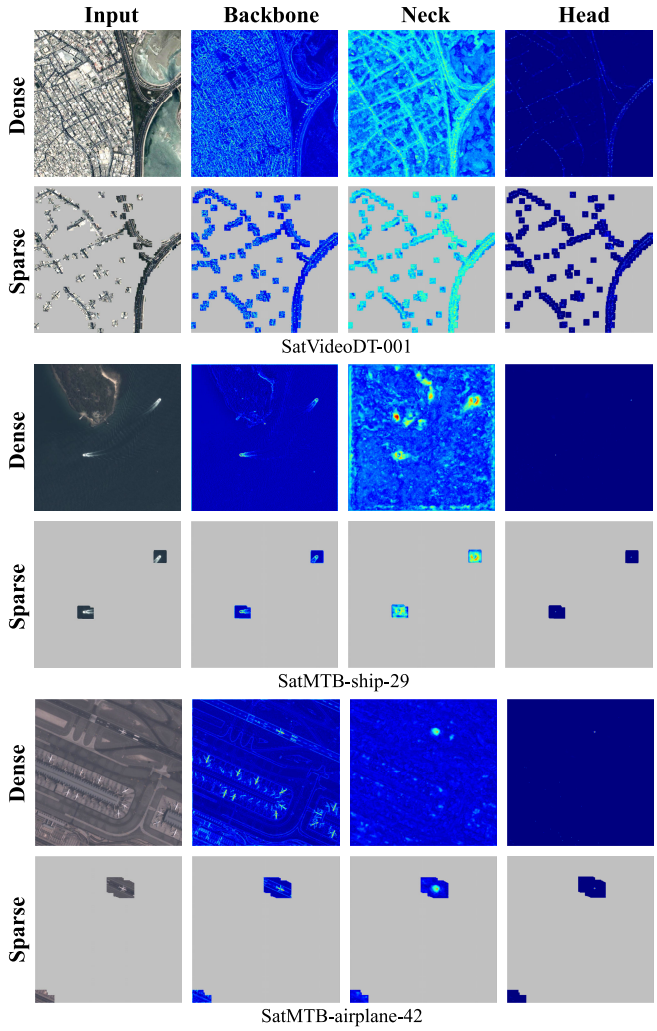


Fig. 10. Comparison of feature maps between the dense and sparse models. Each column (from left to right) represents the model's input and the output feature maps from backbone, neck, and head (Heatmap), which clearly illustrates the fully sparse paradigm of our model. Notably, the sparse feature maps are directly derived from the output of our sparse model. The three components from top to bottom are SatVideoDT-001 [10], SatMTB-ship-29 [6], and SatMTB-airplane-42 [6], respectively.

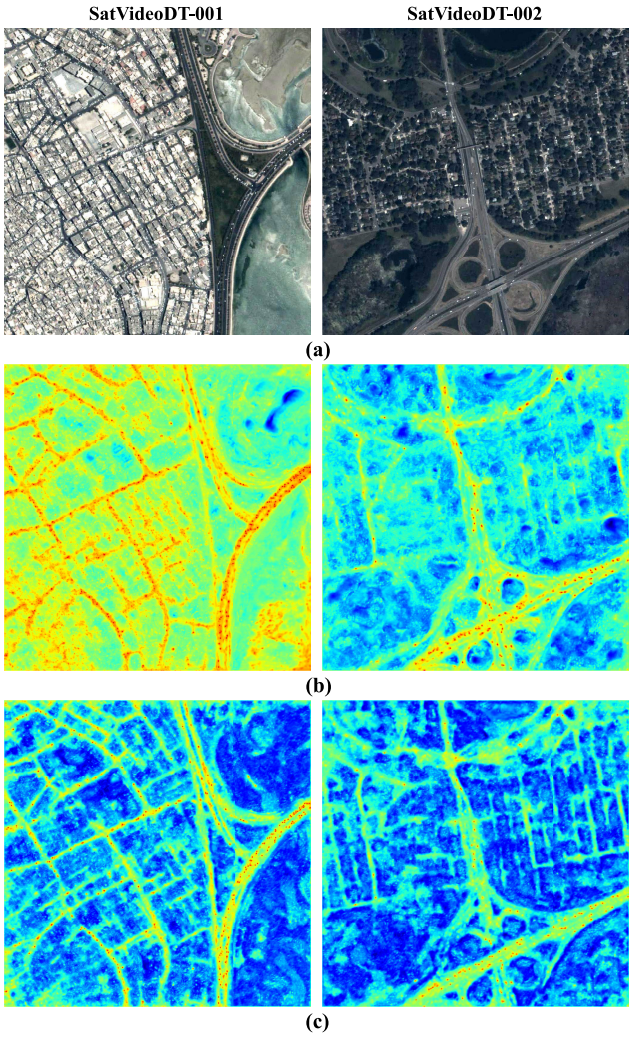


Fig. 11. Visualization of the feature maps for the IAFI. (a) Raw images from SatVideoDT-001 [10] and SatVideoDT-002 [10]. (b) Feature maps before applying IAFI. (c) Feature maps after applying IAFI.

coupled systems through a Gaussian-based modeling approach for small objects and an innovative filtering process. As shown in Table VII, adding GTA results in great improvements, with MOTA reaching 66.2%, IDF1 increasing to 77.6%, and MT rising to 869. Simultaneously, ML decreases to 135 and IDs drops to 275, which demonstrats the effectiveness of GTA in enhancing the accuracy and stability of moving object tracking. In addition, thanks to the plug-and-play nature of GTA, we can apply it to different types of trackers: ByteTrack [17], DSFNet [12], DeepSort [30], Fairmot [18], and CenterTrack [16]. As illustrated in Table VIII, the incorporation of GTA results in consistent performance enhancements across a range of metrics, including MOTA, IDF1, and IDs. Notably, applying GTA to CenterTrack yields the most significant improvement–MOTA increases by 4.0% (from 38.3% to 42.3%), IDF1 rises by 14.6% (from 50.1% to 64.7%), and the number of ID switches decreases by 72.0%. These results indicate that GTA possesses strong generalization ability and

TABLE IX

COMPARISON OF DIFFERENT DATA ASSOCIATION STRATEGY ON SATVIDEODT DATASET [10]. THE BEST RESULTS ARE SHOWN IN BOLD

Association strategy	MOTA ↑	IDF1 ↑	IDs ↓
SORT	23.6%	38.4%	420
BYTE	28.2%	45.6%	281
GTA	**31.0%**	**47.9%**	**277**

can be seamlessly integrated into existing trackers to enhance their performance.

In addition, we compare GTA with two widely used association strategies, SORT [29] and BYTE [17] on the SatVideoDT dataset [10] based on the same detection results. As shown in Table IX, GTA demonstrates clear advantages in terms of MOTA, IDF1, and IDs when compared to the others. Specifically, GTA achieves the highest MOTA of 31.0%, outperforming BYTE (28.2%) and SORT (23.6%), indicating superior overall tracking accuracy

4) Foreground Mask Generation: To enhance tracking stability within the sparse framework, we propose a mask

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TABLE X
ABLATION STUDY OF MASK CONSISTENCY

w/ Consistency	Thresh	MOTA ↑	IDF1 ↑	IDs ↓	FPS ↑
-	-	55.9%	69.9%	330	3.1
✓	0.1	61.6%	**74.8%**	290	2.8
✓	0.2	**62.4%**	**74.8%**	**288**	3.0
✓	0.3	61.1%	73.3%	332	3.0

TABLE XI
ABLATION STUDY OF α FOR FOREGROUND MASK EXTRACTION

α	MOTA ↑	IDF1 ↑	IDs ↓	FPS ↑
1	66.2%	77.6%	275	1.9
3	63.0%	75.5%	277	2.6
5	62.4%	74.8%	288	3.0
7	61.7%	74.0%	303	3.1
9	60.9%	73.4%	297	3.2

TABLE XII
ABLATION STUDY EVALUATING THE ADVANTAGES OF SPARSE MODELS COMPARED WITH DENSE MODELS ON THE SATVIDEODT DATASET [10]

Method	**MOTA** ↑	**IDF1** ↑	FP ↓	FN ↓	MT ↑	ML ↓	IDs ↓	FPS ↑
SPAX (dense)	65.9%	89.6%	24295	71360	**869**	137	278	1.7
SPAX (α1_r20)	**66.2%**	**89.9%**	**23563**	**71198**	**869**	**135**	**275**	**1.9**

consistency strategy, which retains high-activation regions from historical frames for subsequent frame computation. The results of the ablation study are presented in Table X.

By introducing the consistency strategy, we observe a clear performance boost in both MOTA and IDF1 compared to the baseline (no consistency). Specifically, when the consistency threshold is set to 0.2, our method achieves the highest MOTA (62.4%) and IDF1 (74.8%), while maintaining a relatively low number of IDs (288). This configuration strikes the best balance among accuracy, identity consistency, and efficiency, thus motivating our final choice of using consistency with a threshold of 0.2.

In addition, the mask threshold α is a key component that controls the foreground activation ratio of the input images. Therefore, we investigate the influence of the value α on the tracking performance. The quantitative results are shown in Table XI. It can be observed that, as the value of α increases, the foreground activation ratio decreases, leading to an increase in FPS. However, this may also result in the omission of some foreground regions, causing a drop in recall and overall accuracy.

V. DISCUSSION

Our proposed fully sparse framework is training-free and can be directly integrated into any pretrained model for acceleration, offering greater versatility. And the sparse strategy can even filter out certain noisy regions that the deep model may struggle to handle. As shown in Table XII, under the same model weights and configurations, the sparse model (with a mask threshold α of 1 and a radius r of 20) achieves better performance than the conventional dense computation. Furthermore, the hyperparameter activated region radius r offers a flexible tradeoff between computational efficiency and prediction accuracy: decreasing r reduces the activation regions, thereby enhancing computational efficiency; conversely, increasing r helps preserve accuracy. Depending on the varying accuracy and efficiency requirements in real applications, different values of r can be adjusted to achieve the desired balance. In future work, we plan to explore adaptive sparsity mechanisms informed by scale predictions from historical frames, aiming to enhance the model's flexibility in handling the challenges posed by diverse scenarios.

In addition, in this work, we mainly aim to explore the feasibility of a fully sparse paradigm and its effectiveness under the HSF framework for satellite video analysis, which leaves ample room for future research to explore enhancements and customizations, such as extra tracking cues and more sophisticated trajectory modeling techniques [60].

VI. CONCLUSION

This study proposes SPAX for moving object tracking in satellite videos. SPAX addresses the inefficiency caused by background redundancy in the spatial dimension through a fully sparse paradigm and mitigates inaccuracy due to dual-motion coupling in the temporal dimension via HSF, enabling comprehensive exploitation of dynamic information. In addition, GTA significantly enhances the stability and integrity of object trajectories as a plug-and-play strategy. Compared to SOTA trackers, SPAX achieves superior performance in both tracking accuracy and efficiency, demonstrating the impressive capability across different scenes and object categories.

REFERENCES

[1] Q. Yin et al., "Detecting and tracking small and dense moving objects in satellite videos: A benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.
[2] L. Jiao et al., "Transformer meets remote sensing video detection and tracking: A comprehensive survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1–45, 2023.
[3] S. Li et al., "Recent advances in intelligent processing of satellite video: Challenges, methods, and applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6776–6798, 2023.
[4] C. Toth and G. Jóźków, "Remote sensing platforms and sensors: A survey," *ISPRS J. Photogramm. Remote Sens.*, vol. 115, pp. 22–36, May 2016.
[5] Q. He, X. Sun, Z. Yan, B. Li, and K. Fu, "Multi-object tracking in satellite videos with graph-based multitask modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5619513.
[6] S. Li et al., "A multitask benchmark dataset for satellite video: Object detection, tracking, and segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611021.
[7] W. Ao, Y. Fu, X. Hou, and F. Xu, "Needles in a haystack: Tracking city-scale moving vehicles from continuously moving satellite," *IEEE Trans. Image Process.*, vol. 29, pp. 1944–1957, 2020.
[8] J. Zhang, X. Jia, and J. Hu, "Error bounded foreground and background modeling for moving object detection in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2659–2669, Apr. 2020.
[9] T. Yang et al., "Small moving vehicle detection in a satellite video of an urban area," *Sensors*, vol. 16, no. 9, p. 1528, Sep. 2016.
[10] Y. Guo et al., "The first challenge on moving object detection and tracking in satellite videos: Methods and results," in *Proc. ICPR*, 2022, pp. 4981–4988.
[11] J. Feng et al., "Cross-frame keypoint-based and spatial motion information-guided networks for moving vehicle detection and tracking in satellite videos," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 116–130, Jul. 2021.

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[12] C. Xiao et al., "DSFNet: Dynamic and static fusion network for moving object detection in satellite videos," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[13] C. Xiao et al., "Highly efficient and unsupervised framework for moving object detection in satellite videos," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 11532–11539, Dec. 2024.

[14] L. Kong, Z. Yan, Y. Zhang, W. Diao, Z. Zhu, and L. Wang, "CFTracker: Multi-object tracking with cross-frame connections in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611214.

[15] M. Zhao, S. Li, H. Wang, J. Yang, Y. Sun, and Y. Gu, "MP2Net: Mask propagation and motion prediction network for multiobject tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5617515.

[16] X. Zhou, V. Koltun, and P. Krähenbühl, "Tracking objects as points," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 474–490.

[17] Y. Zhang et al., "ByteTrack: Multi-object tracking by associating every detection box," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 1–21.

[18] Y. Zhang, C. Wang, X. Wang, W. Zeng, and W. Liu, "FairMOT: On the fairness of detection and re-identification in multiple object tracking," *Int. J. Comput. Vis.*, vol. 129, no. 11, pp. 3069–3087, Nov. 2021.

[19] R. LaLonde, D. Zhang, and M. Shah, "ClusterNet: Detecting small objects in large scenes by exploiting spatio-temporal information," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4003–4012.

[20] Z. Pi et al., "Very low-resolution moving vehicle detection in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624517.

[21] M. Parger, C. Tang, C. D. Twigg, C. Keskin, R. Wang, and M. Steinberger, "DeltaCNN: End-to-end CNN inference of sparse frame differences in videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 12487–12496.

[22] R. E. Kalman, "A new approach to linear filtering and prediction problems," *J. Basic Eng.*, vol. 82, no. 1, pp. 35–45, Mar. 1960.

[23] C. Li et al., "YOLOv6: A single-stage object detection framework for industrial applications," 2022, *arXiv:2209.02976*.

[24] Z. Ge, S. Liu, F. Wang, Z. Li, and J. Sun, "YOLOX: Exceeding YOLO series in 2021," 2021, *arXiv:2107.08430*.

[25] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," in *Proc. ECCV*, 2024, pp. 1–21.

[26] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[27] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," 2017, *arXiv:1708.02002*.

[28] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[29] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2016, pp. 3464–3468.

[30] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3645–3649.

[31] P. Bergmann, T. Meinhardt, and L. Leal-Taixé, "Tracking without bells and whistles," in *Proc. ICCV*, 2019, pp. 941–951.

[32] Z. Lu, V. Rathod, R. Votel, and J. Huang, "RetinaTrack: Online single stage joint detection and tracking," in *Proc. CVPR*, 2020, pp. 14656–14666.

[33] J. Zhang, X. Jia, J. Hu, and J. Chanussot, "Online structured sparsity-based moving-object detection from satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6420–6433, Sep. 2020.

[34] J. Zhang, X. Jia, J. Hu, and K. Tan, "Moving vehicle detection for remote sensing video surveillance with nonstationary satellite platform," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5185–5198, Sep. 2022.

[35] L. Balzano, R. D. Nowak, and B. Recht, "Online identification and tracking of subspaces from highly incomplete information," in *Proc. Allerton*, May 2010, pp. 704–711.

[36] M. Harandi, C. Sanderson, C. Shen, and B. C. Lovell, "Dictionary learning and sparse coding on Grassmann manifolds: An extrinsic solution," in *Proc. ICCV*, 2013, pp. 3120–3127.

[37] J. He, L. Balzano, and A. Szlam, "Incremental gradient on the Grassmannian for online foreground and background separation in subsampled video," in *Proc. CVPR*, 2012, pp. 1568–1575.

[38] J. Zhang, X. Zhang, Z. Huang, X. Cheng, J. Feng, and L. Jiao, "Bidirectional multiple object tracking based on trajectory criteria in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603714.

[39] W. Zhang, W. Deng, Z. Cui, J. Liu, and L. Jiao, "Object knowledge distillation for joint detection and tracking in satellite videos," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701213.

[40] C. Xu, H. Qi, Y. Zheng, and S. Peng, "Real-time moving vehicle detection in satellite video based on historical differential information and grouping features," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614716.

[41] B. Graham and L. van der Maaten, "Submanifold sparse convolutional networks," 2017, *arXiv:1706.01307*.

[42] Y. Yan, Y. Mao, and B. Li, "SECOND: Sparsely embedded convolutional detection," *Sensors*, vol. 18, no. 10, p. 3337, Oct. 2018.

[43] G. Zhang, J. Chen, G. Gao, J. Li, and X. Hu, "HEDNet: A hierarchical encoder–decoder network for 3D object detection in point clouds," in *Proc. NeurIPS*, 2023, pp. 53076–53089.

[44] G. Zhang, J. Chen, G. Gao, J. Li, S. Liu, and X. Hu, "SAFDNet: A simple and effective network for fully sparse 3D object detection," in *Proc. CVPR*, 2024, pp. 14477–14486.

[45] C. Yang, Z. Huang, and N. Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13658–13667.

[46] B. Du, Y. Huang, J. Chen, and D. Huang, "Adaptive sparse convolutional networks with global context enhancement for faster object detection on drone images," in *Proc. CVPR*, 2023, pp. 13435–13444.

[47] K. Liu et al., "ESOD: Efficient small object detection on high-resolution images," *IEEE Trans. Image Process.*, vol. 34, pp. 183–195, 2025.

[48] Q. Zhou, S. Guo, J. Pan, J. Liang, Z. Xu, and J. Zhou, "PASS: Patch automatic skip scheme for efficient real-time video perception on edge devices," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2023, vol. 37, no. 3, pp. 3787–3795.

[49] Q. Zhou et al., "PASS: Patch automatic skip scheme for efficient on-device video perception," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 5, pp. 3938–3954, May 2024.

[50] W. Lin, T. Chen, and C. Yu, "SpVOS: Efficient video object segmentation with triple sparse convolution," *IEEE Trans. Image Process.*, vol. 32, pp. 5977–5991, 2023.

[51] T. Verelst and T. Tuytelaars, "BlockCopy: High-resolution video processing with block-sparse feature propagation and online policies," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5138–5147.

[52] Y. Li, Q. Hou, Z. Zheng, M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," in *Proc. ICCV*, 2023, pp. 16794–16805.

[53] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[54] F. Yu, D. Wang, E. Shelhamer, and T. Darrell, "Deep layer aggregation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2403–2412.

[55] N. Ballas, L. Yao, C. Pal, and A. Courville, "Delving deeper into convolutional networks for learning video representations," 2015, *arXiv:1511.06432*.

[56] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. CVPR*, 2018, pp. 6848–6856.

[57] C. Xu, J. Wang, Y. Wen, H. Yu, L. Yu, and G.-S. Xia, "RFLA: Gaussian receptive field based label assignment for tiny object detection," in *Proc. ECCV*, 2022, pp. 526–543.

[58] E. Ristani, F. Solera, R. S. Zou, R. Cucchiara, and C. Tomasi, "Performance measures and a data set for multi-target, multi-camera tracking," in *Proc. ECCV*, 2016, pp. 17–35.

[59] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. NeurIPS*, vol. 32, 2019, pp. 8026–8037.

[60] W. Lu, X. Wang, W. An, C. Xiao, Q. Yin, and G. Zhang, "OITrack: Multi-object tracking for small targets in satellite video via online trajectory completion and iterative expansion over union," *Remote Sens.*, vol. 17, no. 12, p. 2042, Jun. 2025.

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